Distributed Generation and Renewable Technologies

Course Work 2

Impact of DGs Penetration on Power System Stability

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Introduction

The integration of distributed generation (DG) into power systems has been increasing rapidly in recent years, as it provides various benefits, such as improved energy efficiency, enhanced power quality, and reduced transmission losses. [1] DGs, including wind turbines and photovoltaic systems, are often connected to the grid through power electronic converters, which enable efficient energy conversion and control of the power flow. However, the increased penetration of DGs can also have significant impacts on the power system, particularly on the stability and security of the grid. [2]

Research in [3] analysed the transient stability of transmission line when applying fault in all branches of the system. The DGs penetration level is increased in the system by two ways; either by loading the DGs with any increase in loading requirements, or by reducing the power generation from centralized plants and cover this shortage by DGs. It has been concluded that implementing DGs in the system improved the transmission line transient stability because when the transmission system is loaded heavier, the connections between centralized generators and loads become weaker, and the generators' oscillations become larger. Inversely, introducing DGs in the power system enhance all of this behaviour. [4]

In this report, we will investigate the effect of penetrating distributed generators on two critical parameters of the synchronous generator: the critical clearing time and the maximum input mechanical power allowed. The critical clearing time is the time interval between a fault in the power system and the point at which the synchronous generator reaches its stable operating point [5], while the maximum input mechanical power allowed is the maximum amount of mechanical power that can be input to the generator without causing instability.

Background theory

In a power system network and under normal operating conditions where the power generated equals the power consumed by the loads the synchronous generators in the systems are operating at a constant mechanical speed. Therefore, the power angle, which is the difference between the generator excitation phase angle and the phase voltage angle of the receiving end (load), remains constant. If a disturbance occurs in the system and this can be in the form of a fault or a power surge in the system, the rotor will then accelerate or decelerate resulting in an unstable power system. The operation of the synchronous generator both in the steady state conditions and in the unstable conditions can be described by the swing equation. The swing equation is given below:

$$M\frac{d^2\delta}{dt^2} = P_m - P_e \qquad [1]$$

where δ is the power angle, M is the angular momentum, P_m is the mechanical power and P_e is the electrical power.

This shows that under steady state operation the mechanical power input should be equal to the electrical power generated. This results in zero net torque applied on the rotor of the generator which results in a constant rotational speed. It can be noted that if the other side of equation is zero that means the power angle δ must be equal to zero. From the swing equation we can see that if a disturbance occurs in the system the rotor will accelerate if the mechanical power input is greater than the electrical power being generated and decelerate when the electrical power is greater than the mechanical power. We can also see that the if the power system is disturbed which in all cases mechanical power will be greater than the electrical power (faults, power surge) the kinetic energy gained during this transient period must be lost for the system to return to a stable state again. This is the basis for the use of the technique used to evaluate the system response after a disturbance has occurred called the equal area criterion. This technique does not require the use of the swing equation to determine stability conditions of the system after disturbance. The stable conditions are determined by equating 2 areas of concern

on the electrical power versus load angle plot. This will be shown below for 2 cases of disturbances in the power network system which are fault and power surge disturbances.

EQUAL AREA CRITERION

POWER SURGE DISTURBANCY

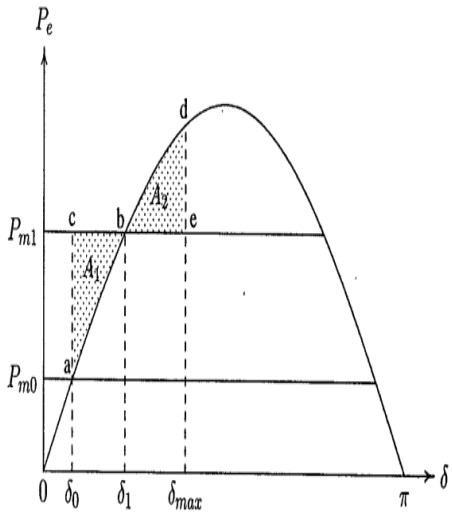

Figure 1.1: Equal area criterion – Power surge

The above plot is given by the power flow equation for a lossless transmission line shown below:

$$P_e = \frac{EV}{X} \sin\delta \qquad [2]$$

where E is the generator excitation voltage, V is the load voltage and X is the transmission or system reactance.

When a power surge occurs the mechanical input power increases to another power value and since changes in practice do not happen suddenly the generator electrical power will start to increase to reach the new value of the mechanical power. During this transient period the rotor is gaining kinetic energy (accelerating) so the rotor speed will increase from the normal steady state rotor speed (w_0). This kinetic energy gained from point a to point b in figure 1.1 is equivalent to the area A_1. In order to return back to steady state operations (w_0) the area above the new mechanical power input must be at the minimum be equal to the area given by A_1. This is because in the area above the new mechanical power (P_{m1}) the electrical power is greater than the mechanical power and therefore the rotor will start to decelerate and lose kinetic energy. Therefore, to return to steady state operation $A_1 must\ equal\ A_2$. Also, during this transient period, the load angle (δ) will move from its initial value of δ_0 to δ_1 and then move to its maximum value of δ_{max}. The load angle will then swing back to δ_1 and oscillate around this value to then subsequently stabilize at this value. This is why δ_1 is called the swing angle. It is also important to note that the total area above the new constant mechanical power input line (P_{m1}) and under the power flow equation (equation 2) is the maximum kinetic energy we can lose, and this is used to determine the amount of power we can increase before the system becomes unstable. By integrating and using equation 2 the solution for the swing angle is given by the following equation:

$$\delta_1 \approx \frac{\delta_0}{2} + \frac{\pi}{4} \qquad [3]$$

FAULT DISTURBANCY

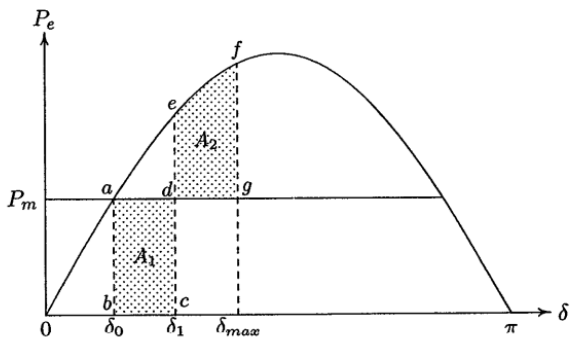

Figure 1.2: Equal area criterion – 3 phase faults at the sending end

The same principle explained in the discussion on using the equal area criterion method for solving for the new steady state operating conditions when a power surge has occurred also apply here for the analysis a fault disturbance. The difference is that during a fault the electrical power to the load goes to zero (is very low) and hence the new area is now given by area A_1 shown in figure 1.2. When a fault occurs the power system operators need to know the time in which the fault should be cleared before the whole system becomes

unstable. Therefore, to determine this, the area which should be equal to area $A_1$is the whole area above the P_m constant line but under the power flow equation (equation 2). Also, the load angle after a fault disturbance should return back to the original load angle before the fault happens. The equations used for obtaining the load angle and the time required for clearing the fault (also known as the critical clearing time) are given below:

$$\delta_2 = \pi - \delta_0$$

$$\delta_1 = \cos^{-1}((\delta_2 - \delta_0)\sin\delta_0 + \cos\delta_2) \quad [4]$$

$$\Delta_{max} = \delta_1 - \delta_0 \quad [5]$$

$$M = J(\frac{2}{P})^2 \times w_e \quad [6]$$

$$H = \frac{M \times w}{2 \times Snom} \quad [7]$$

$$t_{critical} = 2 \times \sqrt{\frac{\delta_{max} \times \sum_i^n S_i H_i}{w_e \times P_{mech}}} \quad [8]$$

where M is the angular momentum, J is the moment of inertia, P is the number of poles, w is the electrical speed in radians per second, S is the apparent power, H is the inertia constant, $t_{critical}$ is the critical clearing time and P_{mech} is the mechanical input power.

Part 1

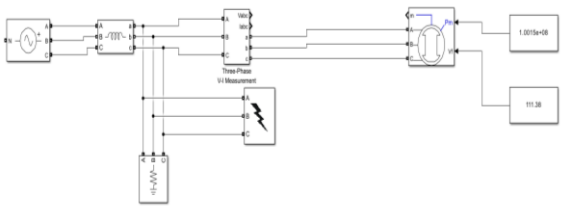

Figure 1: The power system network

$$L = 1.4mH, \quad E = \frac{13.8 \times V_f}{101.3}\ kV \quad [1.1]$$

$$w_s = \frac{2}{P} \times w_e,\ w_e = 2\pi f_e \quad [1.2]$$

L is the synchronous generator inductance, E is the excitation voltage, V_f is the field voltage which in part 1 is equal to 111.38, w_e is the electrical speed in radians per second, f_e = 50Hz.

$$X_s = w_e L = 100\pi \times 1.4mH = 0.4398\Omega \quad [1.3]$$

$$E = \frac{13.8 \times 111.38}{101.3}\ kV = 15.17kV, V = 13.8kV,$$

$$P_{max} = \frac{EV}{X} = \frac{13.8 \times 15.17}{0.4398} MW = 476MW \quad [1.4]$$

The input mechanical power to the synchronous generator derived from the load flow analysis was $100.15MW$ ($P_{mech} = 100.15MW$)

If the power system is stable:

$$P_{mech} = P_e,\ P_e = P_{max}\sin\delta \quad [1.5]$$

δ is the load angle, $\delta = \sin^{-1}\frac{100.15}{476} = 12.146° = 0.21198\ radians$. This is the initial load angle before the fault happens.

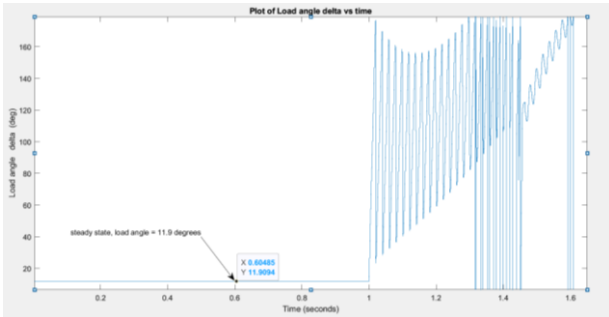

Figure 2: The steady state Load angle

In the above figure it can be seen that the steady state load angle obtained from calculations is equal to the one obtained from simulation. The steady state load angle is less than 90° hence it is a stable operating point of the machine. It should be noted that this steady state angle is way below the 90° point which is very good since any slight disturbances in the power system will not cause the load angle to go beyond 90° and make the system unstable. Therefore, for a more robust power system the load angle should be in the $10° - 20°$ range.

Using the equal area criterion, the new swing angle or new steady state load angle can be obtained from the following equation:

$$\delta_2 = \pi - \delta_0, \quad \delta_1 = \cos^{-1}((\delta_2 - \delta_0)\sin\delta_0 + \cos\delta_2) \quad [1.6]$$

δ_0 is the initial steady state load angle which from the above calculations was 12.146°.

$$\delta_1 = \cos^{-1}((\pi - (2 \times 0.21198)) \sin 0.21198 - \cos 0.21198) = 1.9887\, radians = 113.94°$$

$\delta_{max} = 1.9887\, radians - 0.21198\, radians = 1.77672\, radians$. This is the total swing.

The inertia constant of the machine can be obtained from the following:

$$J = 3.895 \times 10^6\, kgm^2, Snom = 187MVA, H = \frac{M \times w}{2 \times Snom} \quad [1.7]$$

$$P = 40\, poles, w_e = 100\pi, \quad M = J(\frac{2}{P})^2 \times w_e \quad [1.8]$$

$$H = \frac{3.895 \times 10^6 \times \frac{1}{400} (100\pi)^2}{2 \times 187MVA} = 2.56966\, seconds$$

$$t_{critical} = 2 \times \sqrt{\frac{\delta_{max} \times \sum_i^n S_i H_i}{w_e \times P_{mech}}} = 2\sqrt{\frac{1.77672 \times 187 \times 2.56966}{100\pi \times 100.15}} = 0.3295\, s \quad [1.9]$$

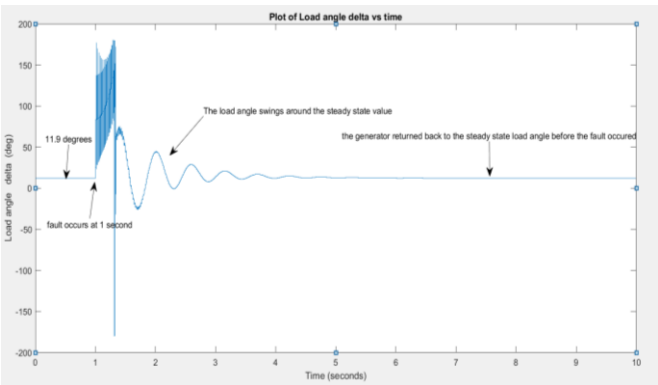

Figure 3: Analysis of power stability after $t_{critical} = 0.3295s$

The above figure shows how power stability is returned after a fault which occurred at a time of 1 second was cleared at a critical clearing time of 0.3295s. The above figure shows that the calculated value is true since stability is maintained. It can also be noted that the load angle will return to the original value of stability before the occurrence of the fault. This shows that this value of the load angle is a stable point and hence why during a power system disturbance the load angle oscillates around this value.

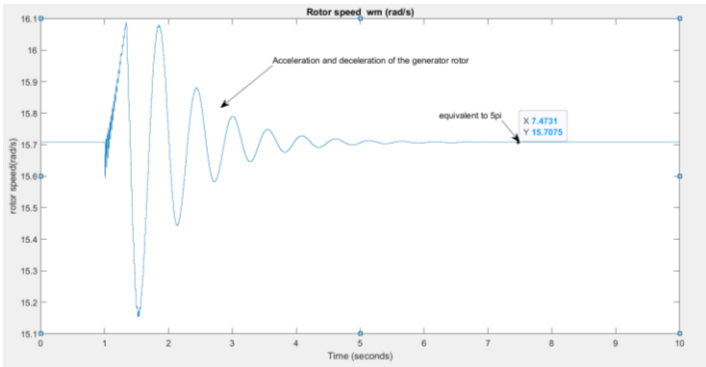

Figure 4: rotor mechanical speed

$$M\frac{d^2\delta}{dt^2} = w_s T_m - w_s T_e \quad [1.10]$$

M is the angular momentum; T represents the torque (mechanical and electrical) and w_s mechanical speed of the rotor (see equation 1.2).

From the above figure it can be seen that our machine operates at a speed of 15.7075 rad/s when in steady state conditions and this is equivalent to the calculated speed of a 20-pole pair synchronous generator operating at 50Hz. Also, when there is a disturbance the generator gains and lose kinetic energy as can be seen from equation 1.10. This results in the machine accelerating and decelerating until it reaches back to the steady state speed when the fault is cleared on time.

Figure 5 below shows the electromagnetic torque during steady state operations and also the value of the electromagnetic torque during a fault until it is cleared. The mechanical power obtained from load flow analysis in the simulation was equal to 100.15MW ($P_m = w_s T_m$) and from figure 5 below we obtain an electromagnetic torque of $6.37746 \times 10^6 Nm$ which if we multiply with w_s (5π rad/s) gives the electrical power.

$$P_e = w_s T_e = 5\pi \frac{rad}{s} \times 6.37746 \times 10^6\ Nm$$
$$= 100.18MW$$

It can be noted that $P_m \approx P_e$ and from equation 1.10 we can see that this is the condition required to maintain a constant speed.

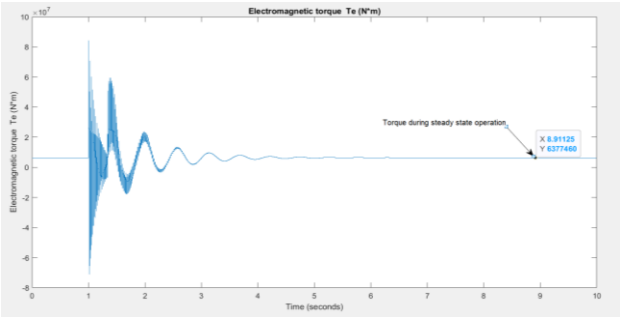

Figure 5: Electromagnetic Torque

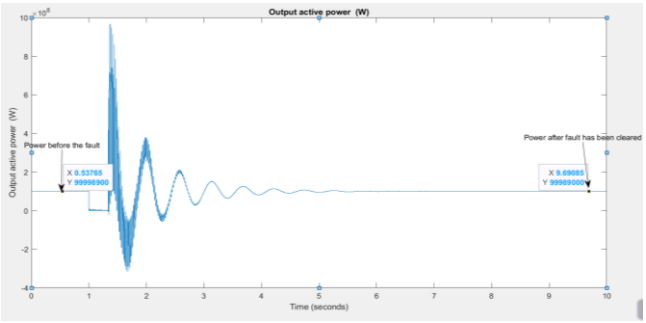

Figure 6: Output electrical power

The output electrical power is equal to 99.989MW which is approximately equal to 100MW the value of the mechanical power generated. It can also be seen that the electrical power output is restored to the original steady state value after the fault is cleared.

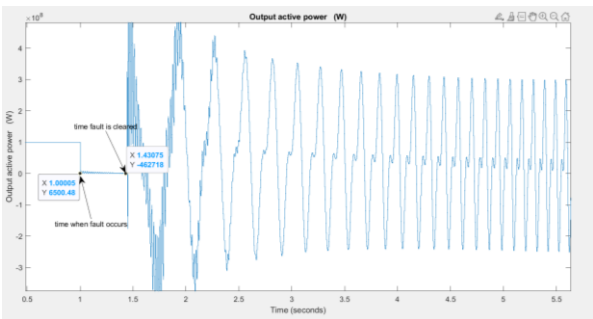

Figure 7: Output Power when a fault cannot be cleared.

The above graph shows that when a fault occurs (t = 1 second) the output electrical power will be zero (blackout) and when the fault was cleared (t = 1.439) the output power will oscillate and if the critical clearing time of the system was exceeded the power will continue to oscillate. Hence, the system is now in a state of power instability that is the system has no power stability. This can result in brown outs and eventually power outages (poor power quality). Comparing the calculated critical clearing time with its simulated value we can see that the calculated value is less than the value obtained from simulation. This is because the equal area criterion which was used to calculate the critical clearing time does not take into account the damper windings and losses in the synchronous machine which are considered in the simulation. Hence, the simulation gives a more accurate value.

$$\%error\ of\ t_{critical} = \frac{0.438s - 0.3295s}{0.3295s} \times 100$$
$$= 32.9\% \qquad [1.11]$$

The above calculation shows that there is a 0.329 error between the calculated value of the critical clearing time which was obtained using the equal area criterion and the value obtained using load flow analysis simulation in MATLAB. So, the trip time of the relay must be set to a value of 0.438s to avoid power outages in the system.

Figures 8 shows that the rotor will continue to accelerate if a fault is not cleared. The rotor speed will increase linearly but in an oscillatory mode. This is because the increase in rotor speed will result in a continuously changing magnetic field in the stator windings which from Lenz's law we know that the stator windings will generate an emf which will create a magnetic field in order to oppose this change in its magnetic field. However, since the mechanical power is always greater than the electrical power generated the machine will continue to increase speed but in an oscillatory manner. This can also be seen from figure 9 which shows that the torque applied on the rotor is also increasing and decreasing periodically.

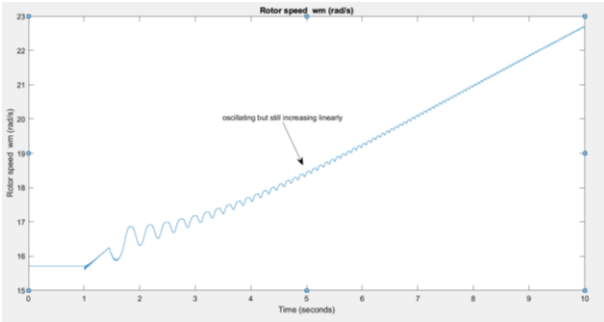

Figure 8: Rotor speed profile when fault cannot be cleared.

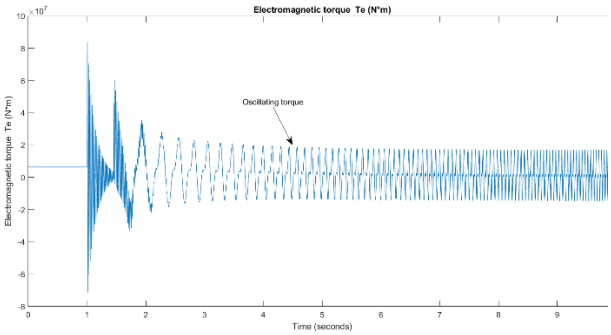

Figure 9: Electromagnetic Torque when fault cannot be cleared.

Part 2

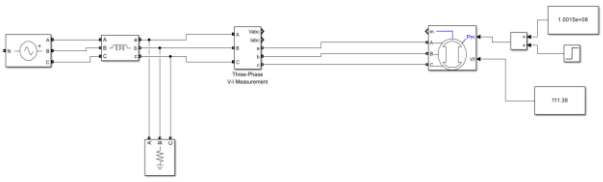

Figure 2.1: Power system network for analysing the effect of a power surge

The initial conditions before a power increase in the power network of figure 2.1 can be obtained from the analysis of the power system network in figure 1 which results in the following initial conditions:

$$\delta_0 = 12.146° = 0.21198\, radians,\ P_{max} = 476MW$$

Using the equal area criterion an equation for obtaining the new steady state load angle is given below:

$$\delta_1 = \frac{\delta_0}{2} + \frac{\pi}{4} = \frac{0.21198}{2} + \frac{\pi}{4} = 0.891388\, radians = 51.07°$$

Therefore, the new value of mechanical power input is:

$$P_{m1} = P_{max} \sin\delta_1 = 476MW \sin 51.07 = 370.3MW$$

$$\Delta P_m = 370.3MW - 100.15MW = 270.15MW$$

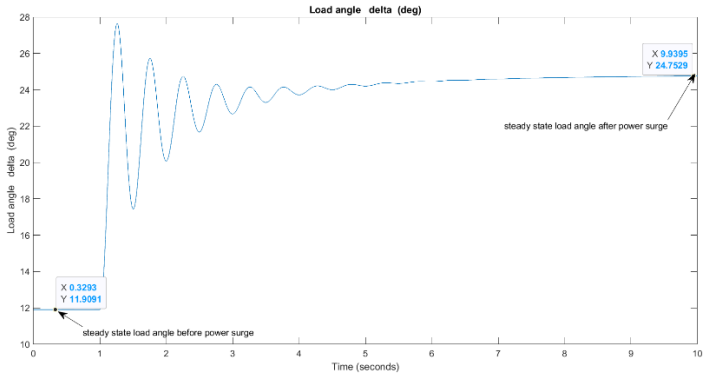

Figure 2.1: Load angle before and after a power surge

The above graph shows that as you increase the power in the system the load angle will increase and operate at a different steady state operating point. Hence, continued increase in the power will lead to power instability. The initial steady state load angle is close to the calculated value of 12.146 degrees. At a power surge of 87MW the calculated load angle will be 23.132 degrees which is close to the one in the above figure (24.7529 degrees).

Figure 2.2 shows that the steady state rotor speed before the power increase is the same as the steady state rotor speed after the power surge. In figure 2.3 we can see that the output power was initially at 100MW during the first 1 second and then raised to 187MW in the steady state part after the power surge disturbance on the system had stabilised. Figure 2.3 also shows that the result obtained from calculation using the equal area criterion does not take into account the rating of the machine and other factors like voltage stability of the system since increased power will result in huge current losses in the system. This in turn will result in the system having voltage instability. To maintain voltage stability the power supplied by the generator should be decreased and then to maintain the same amount of power demanded by the load, distributed generation should be placed close to the users (load).

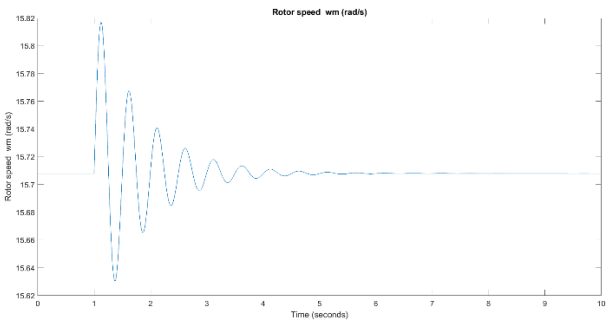

Figure 2.2: Rotor speed

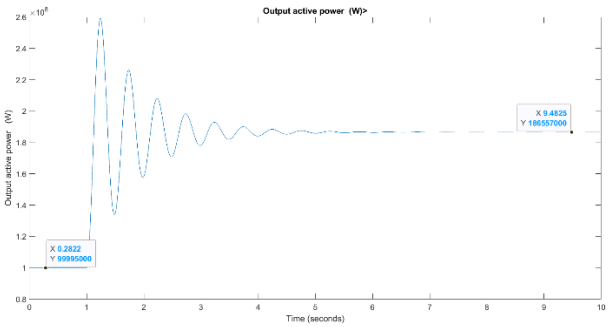

Figure 2.3: Output power

The graph of the electromagnetic torque (figure 2.4) and follows the same pattern as the one for output power this makes sense since power is a product of torque and rotor speed in radians per second. Figure 2.4 also shows that when the mechanical input power increases the value of the torque after the power increase and at the new steady state conditions also increases as compared to its initial value at steady state before the mechanical power increased.

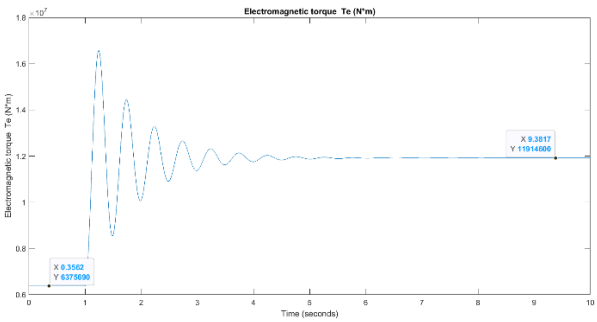

Figure 2.4: Electromagnetic torque

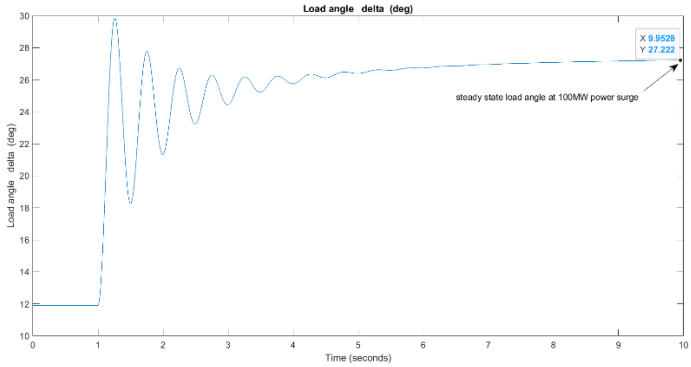

Figure 2.5: Load angle variation during a 100MW power surge

Comparing figure 2.1 and the above figure we can see that increasing the power surge moves the load angle towards the 90-degree point which will eventually lead to an unstable power system. Therefore, even though the value can go beyond the rated power it is not good practice since the equal area criterion has limitations and does not take into account other factors like how this would affect the machine to operate above its rated value.

Since the equal area criterion gives an overestimation of how much power can be increased in the power network system before it becomes unstable and further analysis was considered. The same power system network was then considered and then a fault was introduced into the system to measure the critical clearing time. The power system used is shown in figure 2.6.

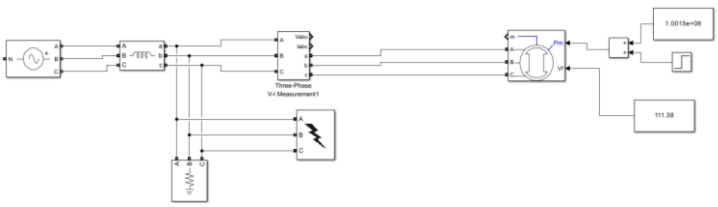

Figure 2.6: Analysis of the impact of a power surge on the critical clearing time

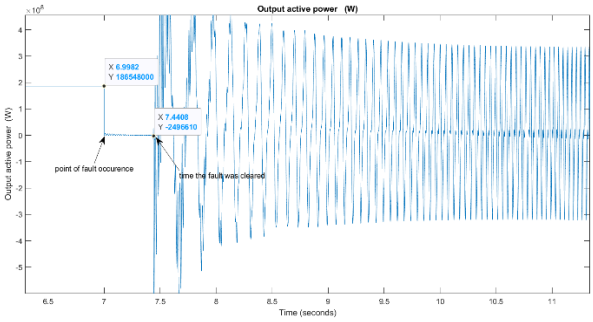

Figure 2.7: Output power for a 187MW power generation at a critical clearing time of 0.438s

Figure 2.7 shows how the output power of the same power system network to that of part 1 but with a difference of an increase of 87MW (187MW total) mechanical input power behaves if we use the same critical clearing time of 0.438s which was enough to stabilize the power system after a fault had occurred. It can be noted that the power system is now unstable. Hence, this shows that increasing the power generated in the power system does not improve power stability.

The power system with a power surge of 87MW has a critical clearing time of 0.245s.

$$\%\, reduction\ of\ t_{critical} = \frac{0.438s - 0.245s}{0.438s} \times 100 = 44.1\%$$

This amounts to a 44 percent reduction in critical clearing time. Therefore, the ideal situation is to increase the power but have value of the critical clearing time close or equal to the original value of the power network system before the power surge.

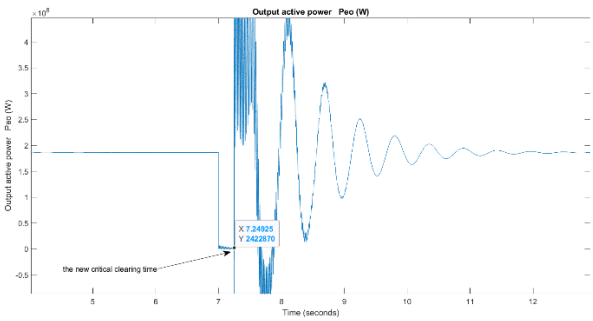

Figure 2.8: Showing the value of the new critical clearing time.

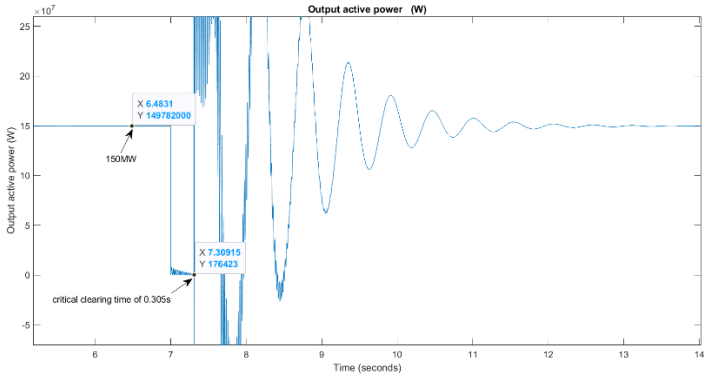

Figure 2.9: The critical clearing time after a 50MW power surge.

The critical clearing time after a 50MW power surge (150MW power system) is now 0.305s this is better than the one for an 87MW power surge but also insuring a significant amount of power increase in the system. It should be noted that the simulated value is very close to the one calculated for a 100MW power network system using the equal area criterion but less than its simulated value of 0.438s. Hence, this 150MW power system is a good compromise between power increase and critical clearing time. The other disadvantage of a power surge is that the fault current will increase and this can be dangerous for people and animals.

Part 3

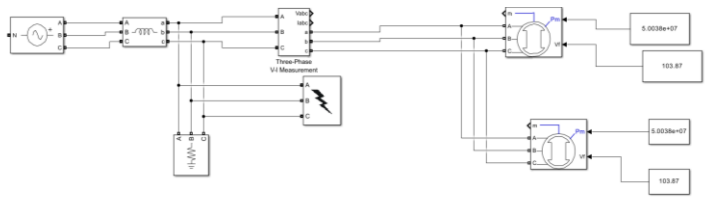

Figure 3.1: The power system network to be analysed.

$$E = \frac{13.8 \times V_f}{101.3}\ kV = 14.15kV, \qquad V_f = 103.87 \quad [3.1]$$

$$V(pu) = \frac{13.8kV}{13.8kV} = 1\ pu\ , E\ (pu) = \frac{14.15kV}{13.8kV} = 1.03\ pu \quad [3.2]$$

$$Z_b = \frac{{V_b}^2}{S_b} = \frac{13.8 \times 13.8}{187} = 1.0184\Omega;\ X(pu) = \frac{0.4398\Omega \times 0.5}{1.0184\Omega} = 0.2159\ pu \qquad [3.3]$$

The value for X (0.4398Ω) was obtained in part 1. The value for X has changed because the two sources are now in parallel.

$$P_{max}(pu) = \frac{EV}{X} = 4.77\,pu \qquad [3.4]$$

$$P_e = \frac{100MW}{187MVA} = 0.535\,pu, \quad P_e = P_{max}\sin\delta \quad [3.5]$$

$$\sin\delta = \frac{0.535}{4.77} = 0.112, \quad \delta_0 = 6.44° = 0.1123\,radians \quad [3.6]$$

$$\delta_2 = \pi - \delta_0, \quad \delta_1 = \cos^{-1}\big((\delta_2 - \delta_0)\sin\delta_0 + \cos\delta_2\big) \qquad [3.7]$$

$$\delta_1 = \cos^{-1}((\pi - (2 \times 0.1123))\sin 0.1123 - \cos 0.1123) = 2.301\,radians = 131.82°$$

$$\delta_{max} = 2.301\,radians - 0.1123 radians = 2.1887\,radians\,,\ H = 2,.56966\,seconds$$

$$t_c = 2 \times \sqrt{\frac{\delta_{max} \times \sum_i^n S_i H_i}{w_e \times P_{mech}}} = 2\sqrt{\frac{2.1887 \times (2.56966 + 2.56966)}{100\pi \times 0.535}} = 0.5174\,s \quad [3.8]$$

Figure 3.1 below shows that if we put a fault clearing time of 0.76 after our last stable clearing time of 0.759s the power system network which has two synchronous generator becomes unstable. Therefore, the maximum clearing time should be 0.759 seconds.

$$\%error\ of\ t_{critical} = \frac{0.759s - 0.5174s}{0.5174s} \times 100 = 46.7\%$$

It can be noticed from the above that the percentage error of the calculated values from the simulated values has increased significantly for this type of power system circuit network with 2 generators. Hence, the equal area criterion does not give the exact values but gives a smaller clearing time than the actual value. Comparing the critical clearing time of part 1 and that of part 3 it can be seen that by increasing the number of generators (generation of electricity) the critical clearing time of the system is improved significantly. The critical clearing time is directly proportional to the number of generators in the system. The aforementioned statement shows the benefit of introducing distributed generation in the power system grid. It should also be noted from the above analysis that the incorporation of distributed generation in the power system will result in a decrease in the overall system impedance. This is very good as a reduced power system impedance will also result in less voltage variations in the system when a disturbance occurs.

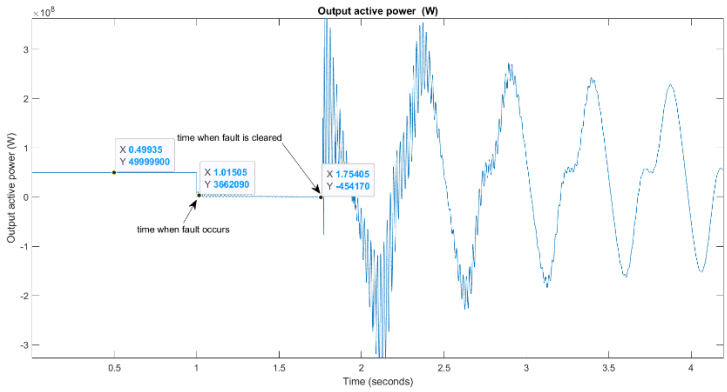

Figure 3.1: Output power the synchronous generator.

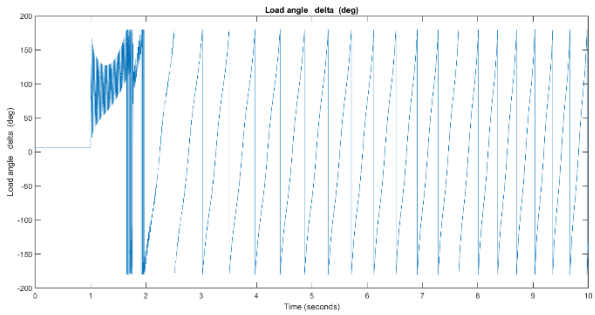

Figure 3.2: Load angle profile after a critical clearing time of 0.76s

The above graph shows that the load angle will oscillate from 0 to 180 degrees which shows that the power system is no longer stable. This is can also be seen in figure 3.3 which shows that the rotor speed to the generator continues to accelerate.

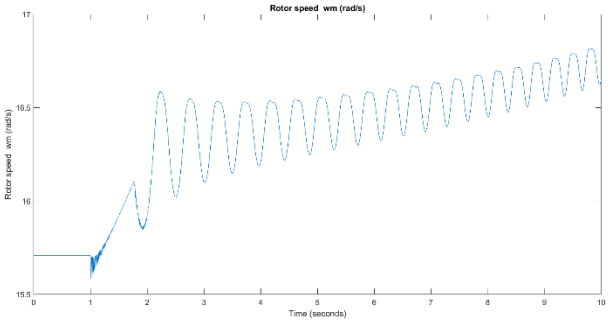

Figure 3.3: Rotor speed after a critical clearing time of 0.76s

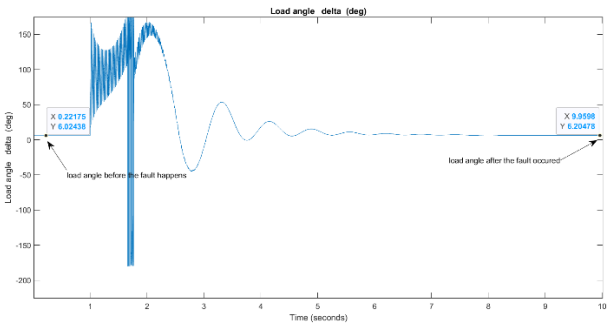

Figure 3.4: Load angle after a critical clearing time of 0.759s

It can be seen that at a critical clearing time of 0.759 seconds stability is returned. The value of the load angle at steady state is equal to 6.024° and this value is equal to the one obtained in equation 3.6 from calculations ($\delta_0 = 6.44°$).

Part 4

The power system network to be analysed is the same as that shown in figure 1.1 but the load is consuming 50MW and the $V_f = 111.73$.

$$V_f = 111.73, \qquad E = \frac{13.8 \times V_f}{101.3}\ kV = 15.22kV \qquad [4.1]$$

$$P_{max} = \frac{EV}{X} = \frac{13.8 \times 15.22}{0.4398} MW = 477.6MW \quad [4.2]$$

The input mechanical power to the synchronous generator derived from the load flow analysis was $100.15MW$ ($P_{mech} = 100.15MW$)

If the power system is stable:

$$P_{mech} = P_e,\ P_e = P_{max} \sin\delta \qquad [4.3]$$

δ is the load angle, $\delta = \sin^{-1}\frac{100.15}{477.6} = 12.104° = 0.21126\ radians$. This is the initial load angle before the fault happens. This is the initial load angle before the fault happens.

$$\delta_1 = \cos^{-1}((\pi - (2 \times 0.21126)) \sin 0.21126 - \cos 0.21126) = 1.9906\ radians = 114.05°$$

$$\delta_{max} = 1.9906\ radians - 0.21126\ radians = 1.77934\ radians$$

$$t_{critical} = 2 \times \sqrt{\frac{\delta_{max} \times \sum_i^n S_i H_i}{w_e \times P_{mech}}} = 2\sqrt{\frac{1.77934 \times 187 \times 2.56966}{100\pi \times 100.15}} = 0.3297\ s$$

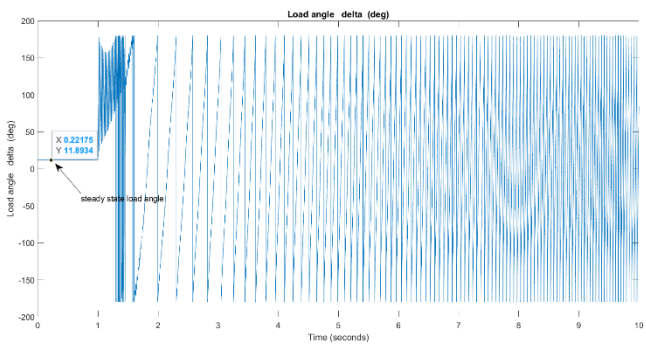

Figure 4.1: Load angle after a critical clearing time of 0.439s

The steady state load angle is close to the one obtained from calculations and is the same as the value obtained in part 1. This because during steady state operations the power being generated by the synchronous generator is the same as the power being generated by the synchronous generator in part 1. Even though the load consumes half of the power as compared to the load in part 1 the other half is consumed at the slack bus in the system to maintain stability.

Figure 4.2 also shows that the electrical output power being generated by the synchronous generator is equal to 100.15MW. It can also be noticed that the output power stabilizes after a critical clearing time of 0.438 second is applied. This is the same as the value obtained in part 1. Hence, this shows that by having a load which consumes less than the generated power the

system will remain stable if it is connected to a slack bus. Also, this system does not improve in terms of power stability when compared to a system which generates the same amount of power but has a load which consumes all the power being generated by the generators. However, this a costly power system since extra infrastructure must be constructed in order to balance the supply and demand to maintain stability. The power system is also not very robust because if the slack bus stops consuming power the generator will be sending too much power to the load which consumes less of the total power being supplied to it.

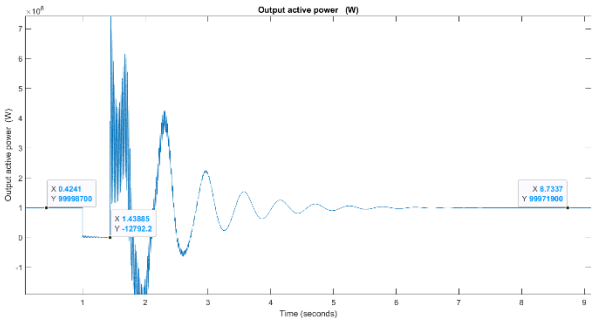

Figure 4.2: Output power after a critical clearing time of 0.438s

But if we match generation and the supply by decreasing the input mechanical power, we can improve the critical clearing time. This can be shown by using the equal area criterion but with input mechanical power equal to 50MW.

$$V_f = 103.87, \qquad E = \frac{13.8 \times V_f}{101.3} \quad [4.4]$$

$$P_{max} = \frac{EV}{X} = \frac{13.8 \times 14.15}{0.4398} MW = 444MW \qquad [4.5]$$

The input mechanical power to the synchronous generator derived from the load flow analysis was $50.038MW$ ($P_{mech} = 50.038MW$)

If the power system is stable:

$$P_{mech} = P_e, \; P_e = P_{max} \sin\delta$$

δ is the load angle, $\delta = \sin^{-1}\frac{50.038}{444} = 6.47° = 0.1129\, radians$. This is the initial load angle before the fault happens. This is the initial load angle before the fault happens.

$$\delta_1 = \cos^{-1}((\pi - (2 \times 0.1129))\sin 0.21129 - \cos 0.1129) = 2.298\, radians = 131.69°$$

$$\delta_{max} = 2.298\, radians - 0.1129\, radians = 2.1856\, radians$$

$$t_{critical} = 2 \times \sqrt{\frac{\delta_{max} \times \sum_i^n S_i H_i}{w_e \times P_{mech}}}$$

$$= 2\sqrt{\frac{2.1856 \times 187 \times 2.56966}{100\pi \times 50.038}}$$

$$= 0.517s$$

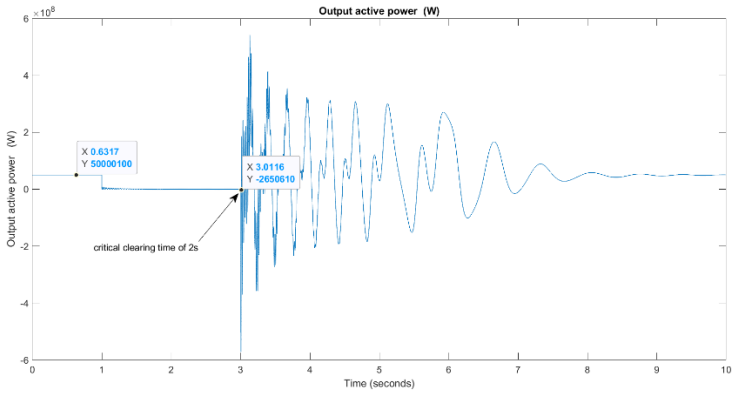

Figure 4.3: Critical clearing time for a 50MW input mechanical power and 50MW load .

The graph shows a significant improvement in the critical clearing if we decrease the amount of power supplied by the synchronous generator by 50%. The critical clearing time was found to 2 seconds. This shows the impact distributed generation would bring since by increasing the amount of generation in the power system we can lessen the amount of power supplied by the synchronous generator in the main power grid. This can also result in increasing the capacity of the distribution network since there will be more power to meet the demand of more users without affecting the power stability of the network.

[4.5] Part 5

The power system network to be analysed is the same as that shown in figure 2.1 but the load is consuming 50MW.

$$\delta_0 = 12.146° = 0.21198\, radians, \; P_{max} = 476MW$$

Using the equal area criterion an equation for obtaining the new steady state load angle is given below:

$$\delta_1 = \frac{\delta_0}{2} + \frac{\pi}{4} = \frac{0.21198}{2} + \frac{\pi}{4}$$
$$= 0.891388\, radians$$
$$= 51.07°$$

Therefore, the new value of mechanical power input is:

$$P_{m1} = P_{max} \sin\delta_1 = 476MW \sin 51.07$$
$$= 370.3MW$$

$$\Delta P_m = 370.3MW - 100.15MW$$
$$= 270.15MW$$

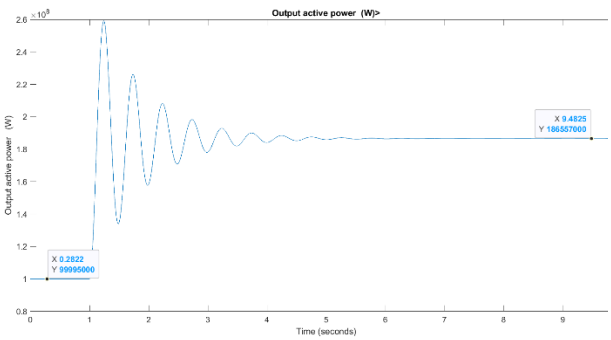

Figure 5.1: Output power for an 87MW power surge

The above output power is the same as the one obtained in part 2. This is because the power generated is the same and though our load is consuming half of the power the other half is being consumed at the slack bus which is also acting as a load.

To further analyse the effect of a power surge on such a power system a fault in this system was introduced and the critical clearing time determined to see if this system is much better than that in part 2.

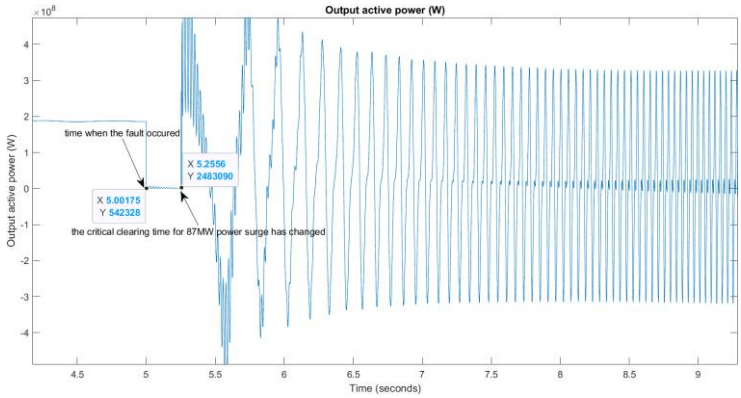

Figure 5.2: The critical clearing time for the new power network at power surge of 87MW

It can be noted that the critical clearing time used for the same power surge of 87MW in part 2 has now change and the power system is no longer stable after clearing the fault at this time. The critical clearing time of this network at a power surge of 87MW obtained from simulation is now equal to 0.2449s a very slight difference from 0.245s. Therefore, the 2 power network systems of part 2 and part 5 behave the same in both a power surge and a fault disturbance.

However, if we decrease the input mechanical power to 50MW to match the load and then analyse the power network under a power surge and a fault disturbance this power system will have an improved critical clearing and also an improved power swing. The following calculations are for an 80MW power surge disturbance when the input mechanical power is 50MW (130MW total).

$$V_f = 103.87, \qquad E = \frac{13.8 \times V_f}{101.3}\, kV = 14.15kV \quad [5.1]$$

$$P_{max} = \frac{EV}{X} = \frac{13.8 \times 14.15}{0.4398} MW = 444MW \quad [5.2]$$

The input mechanical power to the synchronous generator derived from the load flow analysis was $50.038MW$ ($P_{mech} = 50.038MW$)

If the power system is stable:

$$P_{mech} = P_e,\ P_e = P_{max} \sin\delta \qquad [5.3]$$

δ is the load angle, $\delta = \sin^{-1}\frac{130.038}{444} = 17.03° = 0.29724\, radians$. This is the initial

load angle before the fault happens. This is the initial load angle before the fault happens.

$$\delta_1 = \cos^{-1}((\pi - (2 \times 0.29724)) \sin 0.29724 - \cos 0.29724) = 1.783\ radians = 102.13°$$

$$\delta_{max} = 1.783\ radians - 0.29724\ radians = 1.486\ radians$$

$$t_{critical} = 2 \times \sqrt{\frac{\delta_{max} \times \sum_i^n S_i H_i}{w_e \times P_{mech}}} = 2\sqrt{\frac{1.486 \times 187 \times 2.56966}{100\pi \times 130.038}} = 0.264s$$

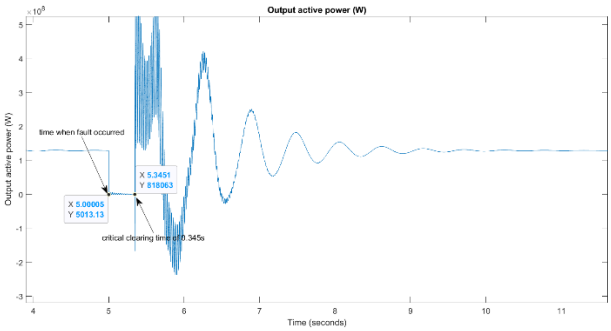

Figure 5.3: The critical clearing time after an 80MW power surge

At a power surge of 80MW the critical clearing time of this power network system is 0.345s which is almost the same value as that of the power system in part with a 50MW power surge. This shows that if we decrease the synchronous generator input mechanical power (decrease its generation capacity) we can add a significant amount of distributed generation in the power system without decreasing its power stability.

Conclusion

Increasing the number of distributed generations in the power system will improve power stability. This is because distributed generation will lessen the system impedance (as seen in part 3) and by improving the critical clearing time. This is due to the fact that a small system impedance will result in a small load angle.

By decreasing the power generated by the synchronous generators in the main grid we can increase the amount of power needed in the grid which will lead to increased number of distribution generation in the power system network.

A power surge disturbance in the power system will decrease the power stability since it will cause the load angle to increase.

The equal area criterion is less reliable when finding the maximum amount of power to be injected into the power grid without affecting stability, but it can be used to estimate the value of the critical clearing time needed in the power system network.

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